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Advancements in Spatio-temporal agricultural drought monitoring and modeling: a comprehensive review on multi-source remote sensing and machine learning techniques

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Abstract

Agricultural drought occurs when soil moisture is insufficient to meet the specific requirements of a given crop during its growth stage, resulting in reduced crop health and yield. Advanced geospatial techniques can effectively monitor and model the spatiotemporal dynamics of agricultural droughts using multiple satellite data products and machine learning (ML) methods, which are accurate, cost-effective, and transferable across space and time. In this study, we comprehensively synthesize the contributions of multi-source remote sensing data and artificial intelligence (AI) approaches (including machine learning and deep learning (DL)) for monitoring, evaluating, and modelling agricultural drought. Additionally, we comprehensively reviewed the latest developments in time-series satellite data analysis, remote-sensing spectral indices, and numerical models used worldwide for rapid, accurate drought monitoring and modeling. Moreover, we critically examined several case studies of drought assessment using ML/DL coupled with satellite data, multi-parametric datasets, and ground-based data to monitor agricultural droughts at local and regional scales and for prediction modelling, owing to their ability to handle complex, non-linear, and large-scale datasets effectively. Overall, big data analytics has great potential for monitoring and assessing agricultural drought using multi-source satellite data, combined with ensemble, hybrid, and physics-based machine learning and deep learning approaches. The review offers a solid theoretical framework and a comprehensive summary of advanced geospatial and ML approaches used for agricultural drought monitoring and modeling, which can be utilized by decision-makers, policy planners, and research organizations to inform policies related to sustainable and climate-smart agriculture.

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1 Introduction

The drought is considered one of the most complex natural catastrophes since it is difficult to identify its beginning, duration, severity, and geographic coverage (Aadhar and Mishra 2017). Such an unpredictable nature causes an adverse impact on the terrestrial ecosystem, consequently impeding its achievement of the Sustainable Development Goals (SDGs). The large-scale climate models indicate that the frequency of occurrence of drought would substantially increase in the 21st century (Holgate et al. 2023). Specifically, under the high emission scenario, the compound drought and heat wave events are expected to increase significantly in South Asia, and the population in 70% of the landmass area of the region is going to be affected by such extreme events by the middle of the 21st century (Ullah et al. 2023). Drought is a stochastic natural and climatic disaster that occurs frequently in many parts

of the world at any time without any indication (Dutta 2018). Droughts can be defined as an absence or deficiency of precipitation that does not meet demand, and they typically occur over a period of time, such as a week, months, or even years. It is influenced by different factors such as high temperatures, low humidity, strong winds, duration, timing, severity, and distribution of rainy days (Mishra and Singh 2010). It results in economic loss, affecting the largest part of society, agriculture, and environmental problems (Halwatura et al. 2017). It is estimated that drought led to 11.73 million casualties between 1900 and 2021. Further, an economic loss of 6 to 8 billion USD is estimated per year, which is far ahead of any natural disaster (Xu et al. 2023). According to the Australian Bureau of Agricultural and Resource Economics, the 2006 drought had a devastating impact on rural Australia. Winter cereal production decreased by 36%, resulting in estimated economic losses of around AUD \$3.5 billion. This significant downturn put many farmers under serious financial pressure, threatening the stability of farming communities across the country (Mishra and Singh 2010; Wong et al. 2010). It is evident from recent severe droughts in around the world such as northern China (2010–2011), California (2011–2017), Europe (2011, 2015, 2018), the Horn of Africa (2011–2012), the Caribbean (2013–2016), southeastern Brazil (2014–2017), South Africa (2015–2016, 2018), India (2016, 2019), and Vietnam (2016) that the degree of exposure, susceptibility, and lack of coping capacity is linked to the risk of adverse effects associated with droughts (Meza et al. 2020). Drought monitoring is a crucial aspect of planning, mitigation, and policymaking for sustainable community development.

There are numerous definitions of drought. The World Meteorological Organization (WMO) defines drought as a persistent natural hazard characterized by lower-than-expected or lower-than-normal precipitation that, when prolonged over a season or longer period, is insufficient to meet the demands of human activities and the environment. (WMO 2006). According to the WHO, the drought can be classified into four categories (Fig. 1), namely, (i) Meteorological drought, (ii) Hydrological drought, (iii) Agricultural drought, and (iv) Socio-economic drought, based on their induced factors.

1.1 Meteorological drought

Meteorological drought is a climatic anomaly that refers to the deficiency of precipitation over a certain period, significantly affecting soil-water-crop balances (Ayugi et al. 2022). It could be linked to different variables such as high temperatures, low humidity levels, and high evapotranspiration rates, all of which could potentially exacerbate the

consequences of meteorological drought (Dingman 2002). Indian Meteorological Department (IMD) further classified this into three classes: (i) Mild meteorological drought (when seasonal rainfall over an area is less than 25% of the normal rainfall), (ii) Moderate meteorological drought (rainfall in between 26 and 50%), and (iii) Severe meteorological drought (rainfall deficit < 75%). Researchers used numerous techniques or indices to understand and monitor the condition of meteorological drought, including the Standard Precipitation Index (SPI), Rainfall Anomaly Index (RAI), and Palmer Drought Severity Index (PDSI) (Alley 1984; Aryal et al. 2022).

1.2 Hydrological drought

Hydrological drought can be defined as a condition where the surface and subsurface water resources are inadequate to meet the normal and specific needs in a specific area (Van Loon 2015). It occurred due to a lack of precipitation and overuse of water from reserves such as rivers, reservoirs, and aquifers. Hydrological drought reduces streamflow, lowers groundwater levels, and can lead to the drying of lakes, wetlands, and rivers. These impacts are often intensified by higher temperatures and increased evaporative demand, which accelerate water loss from both soil and open water bodies. As a consequence, hydrological drought strongly affects aquatic ecosystems by shrinking habitat availability, degrading water quality, and increasing ecological stress on freshwater organisms (Lake 2011).

1.3 Agricultural drought

Agricultural drought is connected to meteorological and hydrological drought (Wang et al. 2016). The rainfall deficiency and water scarcity result in reduced soil moisture content, which affects plant growth and leads to a decline in crop production, a phenomenon referred to as agricultural drought (Seleiman et al. 2021). Soil moisture content is directly related to rainfall or water. If rainfall is deficient, the soil moisture in the area also decreases, negatively impacting crop growth and production, and affecting the country's food security. Various factors, including crop variety, development stage, soil type, weather, and plant water requirements, determine the amount of water needed for agriculture (Mishra and Singh 2010). Agricultural drought can occur at any time during the crop's growth. The need for soil moisture varies with crops; hence, the effects of agricultural drought will differ across regions depending on the crops grown (Chowdhury and Gore 1989). It can be further classified into five categories: early-season drought, mid-season drought, terminal season drought, permanent drought, and apparent drought (Sai et al. 2016).

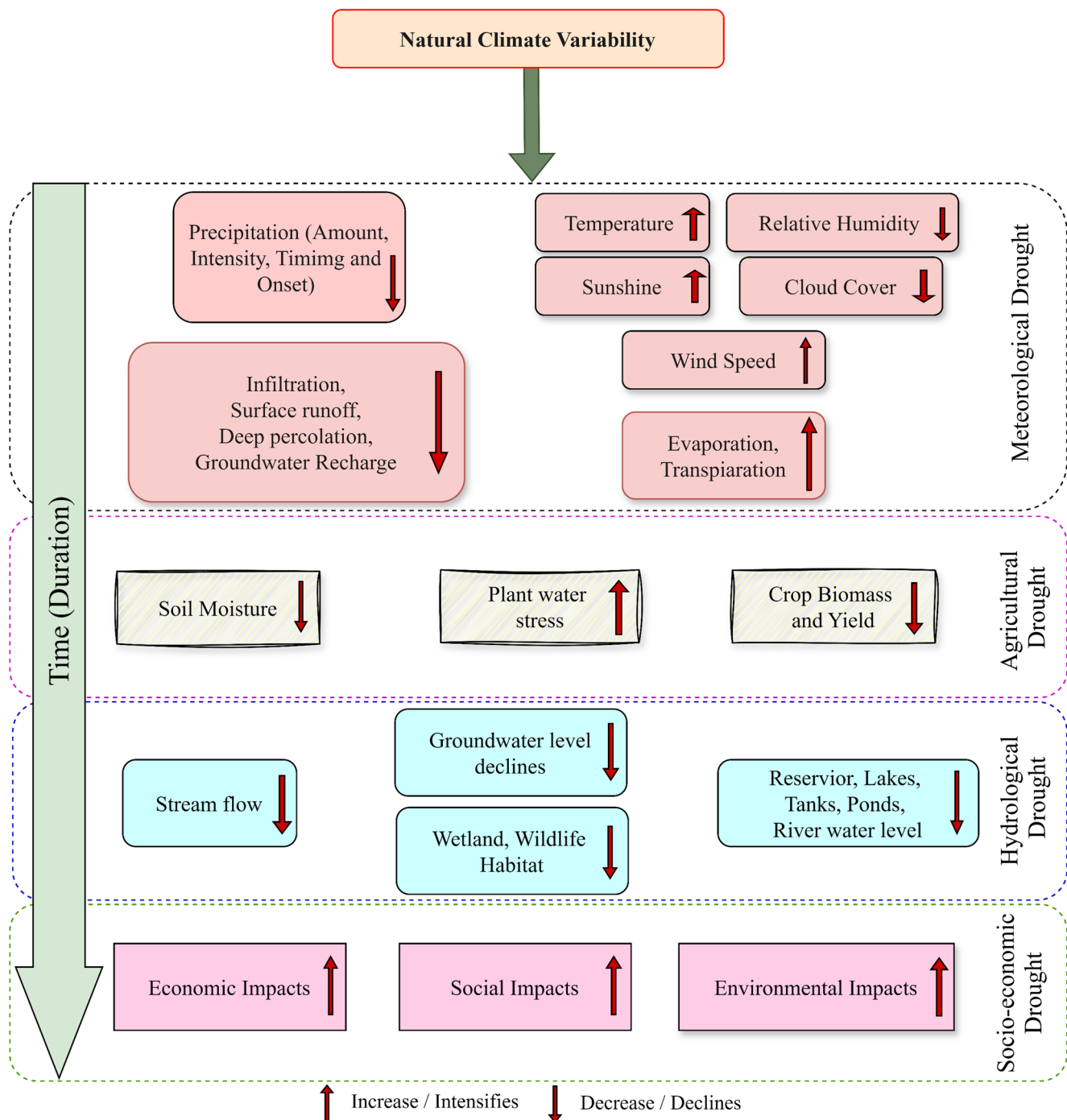


Fig. 1 The sequence of drought impacts associated with meteorological, hydrological, and agricultural drought (adapted and modified from NDMC, 2012)

1.4 Socio-economic drought

Water scarcity refers to the condition in which the available water cannot meet the supply and demand of both human society and animals. This affects health, the economy, society, and quality of life (Zhao et al. 2019;

Ziolkowska 2016). Socioeconomic drought primarily concerns the social dimensions, emphasizing how drought affects and interacts with human activities. It focuses on the extent to which the environment and society can develop sustainably with the help of the available water resources. (Wang et al. 2022).

Our primary objectives are to review the applications of satellite remote sensing and Artificial Intelligence (AI) based on geospatial modeling techniques for characterizing and predicting agricultural drought conditions, including Machine Learning (ML) and Deep Learning (DL) approaches. This review aims to provide a comprehensive overview of recent trends in agricultural drought modeling studies that integrate machine learning and deep learning with big data concepts, using multi-source remote sensing and climatic indices.

2 Methodology

A literature search was conducted using Google Scholar, BASE, and Semantic Scholar (searches performed up to August 2025). The search focused on two complementary keyword sets: (1) (“Drought” OR “Agricultural drought”) AND (“remote sensing” OR “GIS”), and (2) (“Drought” OR “Agricultural drought”) AND (“AI” OR “machine learning” OR “ML”). These searches were conducted separately to achieve maximum coverage of the literature from both the Remote Sensing/GIS and ML/DL perspectives. There were no restrictions on publication year or study location in the search process to ensure the inclusion of both foundational and recent contributions to drought monitoring and modeling. The final set of eligible studies spanned the period from 1974 to 2025. As the keyword search returned a very large number of records, a relevance-ranked screening strategy was employed, focusing only on those papers that met the eligibility criteria. For each keyword query, results were sorted by relevance, and the top 300 records were screened (Haddaway et al. 2015). Additional relevant studies were identified through citation snowballing. Studies were eligible for inclusion if they addressed drought or agricultural drought from the perspective of Remote Sensing/GIS and/or ML/DL approaches. Whereas predefined search keywords were used for the initial retrieval of candidate records, and the final inclusion was determined through title/abstract screening, as well as full-text assessment (including methodological details). In addition, foundational methodological studies (e.g., drought indices, remote sensing indices, and core ML algorithms) were included when they were directly relevant and frequently cited by eligible drought studies. Papers were excluded if they dealt with unrelated ecological or engineering topics, did not provide a detailed methodology, were in a non-English language, or represented the same paper through multiple databases. The final dataset comprised 205 publications, including journal articles, Proceedings Papers, Early Access Articles, and Conference Papers. From the selected studies, the objectives, study area and scale,

drought type and indices, data sources, and sensors (for RS/GIS), as well as the algorithms and features used (for AI studies) were systematically collected (Table 1).

3 Impact of agricultural drought

The impact of agricultural drought can vary over time. It adversely affects various aspects of society, their economic condition, and the environment of the concerned area (Wilhite and Glantz 1985). The growth of plants primarily depends on soil moisture and nutrient availability. Factors such as water supply, insect infestations, and plant diseases can negatively affect plant growth (Dixon 2015), leading to reduced crop quality and yield. When plant growth is affected, it also impacts livestock production, which ultimately results in economic losses, displacement of people, and a risk of famine, as well as public health issues, habitat damage, and an increase in farmers’ suicides (Sundararajan et al. 2021). Figure 2 illustrates the major areas of drought vulnerability worldwide, showing that large-scale, intensive drought events have occurred, affecting extensive areas of Europe, Africa, Asia, parts of Australia, South America, and Central America.

4 Rationale for agricultural drought monitoring and modeling

Agricultural drought monitoring involves the systematic observation of soil moisture, crop conditions, and related hydroclimatic variables to detect emerging water stress and assess its impacts on crop production (Ajaz et al. 2019; Kaliraj et al. 2024). Unlike flood events, drought processes develop slowly and remain challenging to detect in their early stages, while climate change further increases the risks for climate-sensitive sectors, particularly agriculture (Mishra and Singh 2010). Agricultural drought modeling supports monitoring by enabling early warning, identifying vulnerable areas, and guiding interventions that reduce impacts on food production and water resources (Solh and Ginkel 2014; Ginkel and Biradar 2021). Since a large fraction of the global population directly depends on agriculture, severe drought impacts can trigger cascading socio-economic consequences, including livestock loss, food insecurity, and increased burden on government resources (Bodner et al. 2015; Mera 2018; Speranza et al. 2008). Therefore, timely and reliable drought monitoring and modeling are essential for providing policymakers, the Government, farmers, and other stakeholders with accurate information on cropping strategies, irrigation management, and food security planning.

Table 1 Summarized workflow for the literature search, screening, and selection based on eligibility criteria

Step	Stage	Description
1	Databases/search engines	Google Scholar, BASE, and Semantic Scholar
2	Search period (coverage)	Searches performed up to August 2025; eligible studies span 1974–2025
3	Keyword set 1	(“Drought” OR “Agricultural drought”) AND (“remote sensing” OR “GIS”)
4	Keyword set 2	(“Drought” OR “Agricultural drought”) AND (“AI” OR “machine learning” OR “ML”)
5	Restrictions applied	No restriction on publication year or study region; English-language only
6	Initial retrieval	Keyword searches returned a very large number of results across search engines
7	Screening approach	A relevance-ranked screening strategy was applied to manage the large volume of records
8	Screening threshold	For each keyword query, results were sorted by relevance, and the top 300 records were screened
9	Duplicate handling	Duplicate records across the three sources were removed
10	Eligibility assessment	Screening was conducted sequentially: title → abstract → full text
11	Additional records	Additional eligible studies (including foundational and highly cited papers) related to drought were identified through citation snowballing.
12	Inclusion criteria	Studies were included if they: (i) focused on drought/agricultural drought, and (ii) applied RS/GIS and/or AI/ML/DL approaches, and (iii) provided sufficient methodological details for extraction (indices/sensors/data sources/algorithms/features).
13	Exclusion criteria	Studies were excluded if they: (i) were not drought/agricultural drought focused, (ii) lacked RS/GIS or AI/ML/DL methodological relevance, (iii) provided insufficient methodological details, (iv) were non-English, or (v) were duplicates
14	Documents type	Eligible document types included journal articles, proceedings papers, early access articles, and conference papers
15	Final included studies	205 publications were included in this study

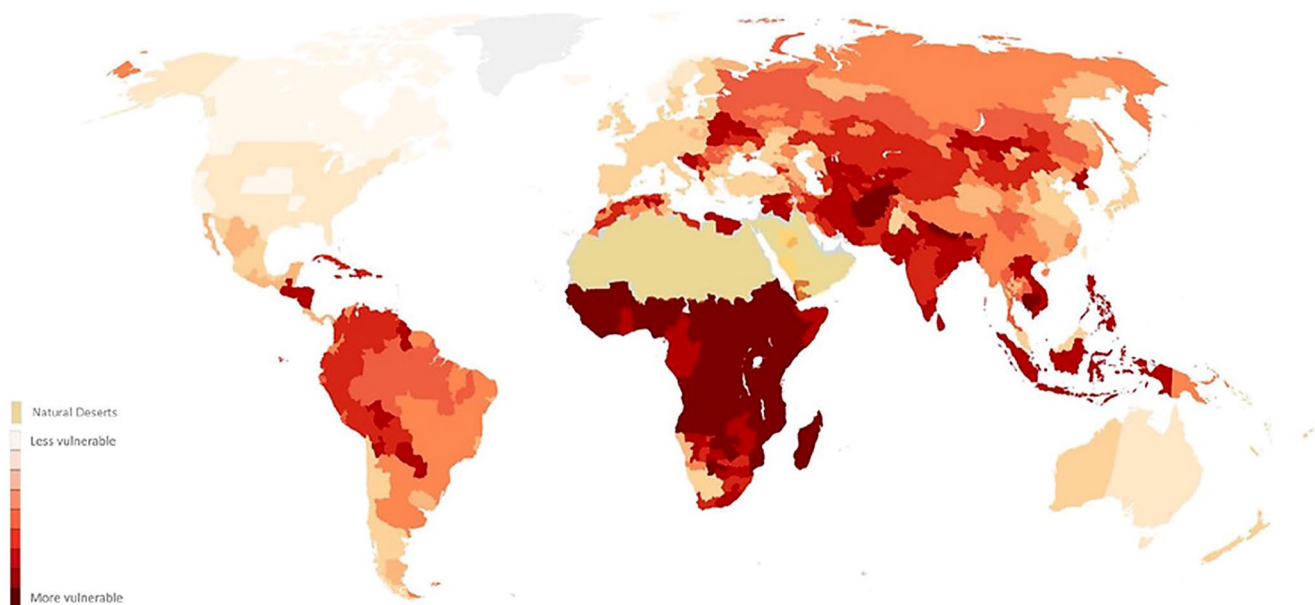
**Fig. 2** Drought vulnerability map of the world. (Source: UNCCD. Global Drought Snapshot, 2023)

Table 2 List of regional and global drought monitoring platforms (collected and modified from: Alahacoon and Edirisinghe 2022)

Drought Monitoring Platform	Drought Monitoring Indices	Source/References
ClimatView - a tool for viewing monthly climate data	SPI	(Japan Meteorological Agency n.d.)
Flood and drought portal	SPI, Effective Drought Index (EDI), Normalized Difference Vegetation Index (NDVI), Vegetation Condition Index (VCI), Soil Water Index (SWI), Vegetation Health Index (VHI), Agricultural Stress Index (ASI), Combined Drought Index (CDI)	(UNEP-DHI n.d.)
North American Drought Monitoring System	SPI, Percent of Average Precipitation, PDSI	(NOAA-NCEI n.d.)
Global Drought Information System	SPI, VHI	(NOAA-GDIS n.d.)
Global integrated drought monitoring and prediction system (GIDMaPS)	SPI, Standardized Soil Moisture Index (SSMI), Multivariate Standardized Drought Index (MSDI)	(Hao et al. 2014)
IRI Global drought analysis tool	SPI	(IRI-Columbia University n.d.)
SPEI Global Drought Monitor	Standardized Precipitation Evapotranspiration Index (SPEI)	(CSIC n.d.)
African Flood and Drought Monitoring System	SPI, NDVI	(AFDM-Princeton University n.d.)
CIIFEN Drought Monitor		(CIIFEN n.d.)
European Drought Observatory	CDI, SPI, Soil Moisture Anomaly (SMA), Vegetation Productivity Anomaly (fAPAR Anomaly)	(EC & JRC n.d.)
South Asia Drought Monitoring System (SADMS)	Integrated Drought Severity Index (IDSI), SPI, SMI, VCI, Temperature Condition Index (TCI), Precipitation Condition Index (PCI)	(IWMI n.d.)

However, drought monitoring based only on station observations is often insufficient due to sparse spatial coverage and limited representativeness, particularly in data-scarce regions (Mishra and Singh 2010; Sardar et al. 2021). Satellite remote sensing can partially address these limitations by providing spatially continuous and near-real-time information on vegetation condition, land surface temperature, and surface moisture anomalies (AghaKouchak et al. 2015; Ahady et al. 2025; Łagiewska and Bartold 2025; Mullapudi et al. 2023). Nevertheless, remote sensing products can be affected by cloud contamination, index saturation in dense canopies, and differences in sensor resolution (Jiao et al. 2021; Qin et al. 2021). Monitoring agricultural drought requires integration of multiple sources of information to complement precipitation data (Fioravanti et al. 2025). While ML and DL approaches are increasingly being adopted to integrate multi-source datasets and capture non-linear relationships between drought, climate, and vegetation drivers (Hou et al. 2025; Obsie and Liu 2025; Prodhan et al. 2022; Rahmati et al. 2020; Rao et al. 2025; Senapati et al. 2025; Tanriverdi and Batmaz 2025). Still, their performance may not transfer well across agroclimatic regions, and many models remain difficult to interpret or validate due to limited ground observations (Houmma et al. 2022; Mi et al. 2025; Prodhan et al. 2022). Therefore, a comprehensive synthesis of remote sensing indices and sensors, along with an AI framework that includes their strengths,

limitations, and suitability across scales, is essential to support improved drought early warning and risk reduction. In this context, Table 2; Fig. 3 summarize the major regional and global drought monitoring and forecasting platforms.

5 Role of remotely sensed data in understanding the agricultural drought

Remote sensing is highly effective for characterizing spatio-temporal changes in land cover in terms of essential physical properties, such as surface radiance and emissivity data (Orhan et al. 2014). Compared with traditional datasets, such as rainfall records and in situ observations, remote sensing provides operational, sustained, and near-real-time measurements with broad spatiotemporal coverage and consistency, thereby overcoming the limitations of measurement error, spatial sparsity, and reporting delays (Jiao et al. 2021; Chandramohan et al. 2024). Since the late 1980s, remote sensing has been utilized for drought monitoring (Tucker and Choudhury 1987) (Fig. 4). This approach provides data with time intervals and historical records for the same location over a longer period (Stagge et al. 2015). Drought can be assessed using Earth observation (EO) satellite systems, which broadly include: (i) meteorological satellites, such as METEOSAT and NOAA/AVHRR, which provide frequent (near-real-time) observations useful for capturing

Fig. 3 Regional and global drought monitoring and prediction systems around the world (adapted and modified from Gyaneshwar et al. 2023)

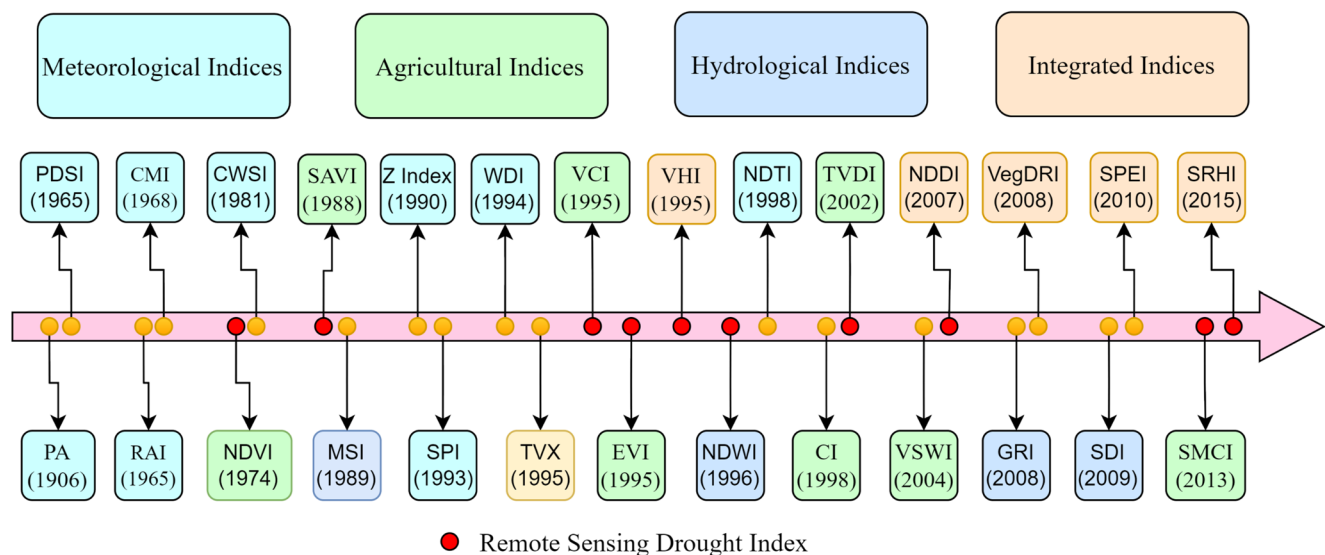
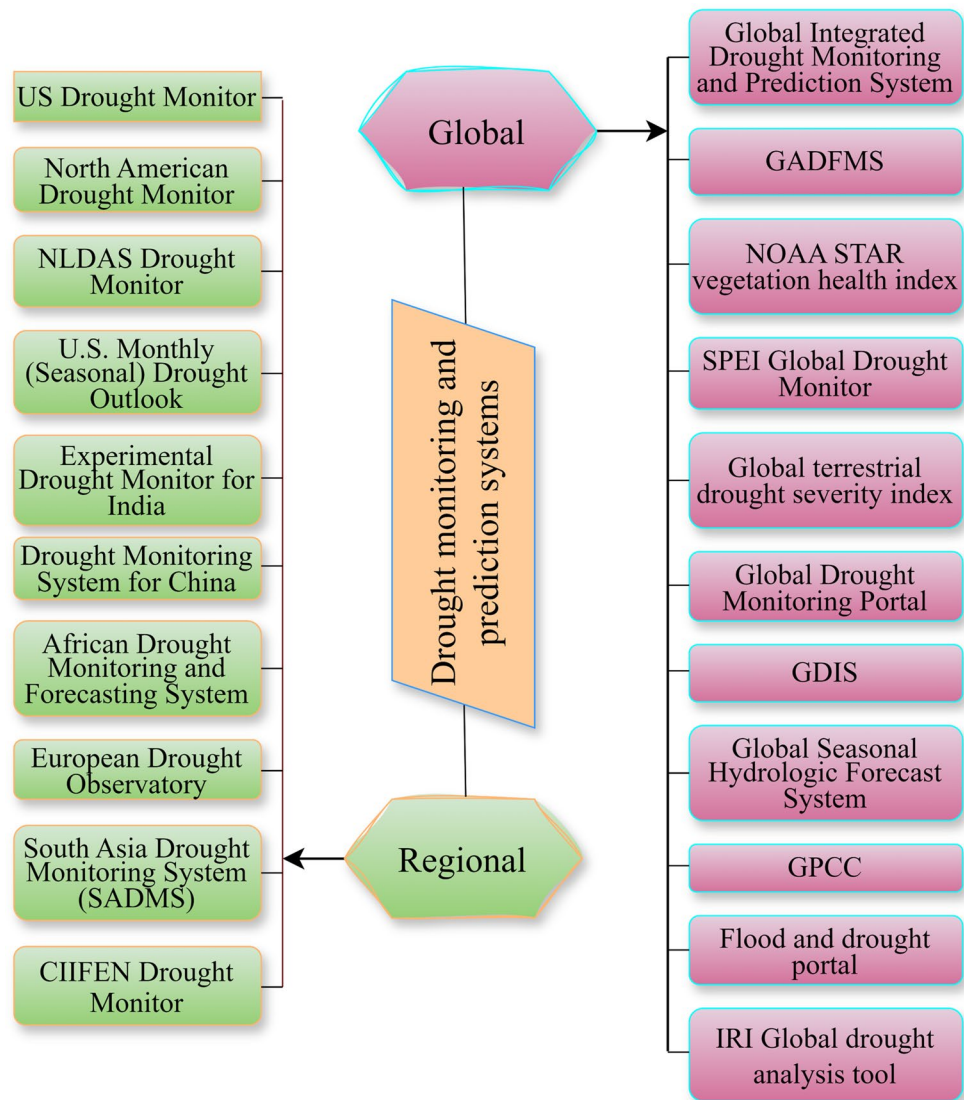


Fig. 4 Development process of drought monitoring indices (adapted and modified from Liu et al. 2016)

large-scale atmospheric and vegetation anomalies related to drought (Dalezios et al. 2009), and (ii) land EO satellites, such as the Landsat and Sentinel missions, as well as commercial high-resolution sensors (e.g., SPOT, IKONOS, WorldView), which provide detailed information on vegetation condition, land surface characteristics, and land cover changes relevant to agricultural drought assessment (Gaikwad et al. 2022).

Large amounts of satellite data are available for agricultural drought research, including the Landsat and Sentinel series, the Moderate Resolution Imaging Spectroradiometer (MODIS), and Soil Moisture Active Passive (SMAP), which are often used for drought monitoring (Table 3). These all provide near-real-time, medium to high-resolution datasets for assessing vegetation cover, soil moisture, crop condition, and water body estimation for lakes, rivers, and reservoirs (Prodhan et al. 2022). Landsat series' archived

and real-time data is one of the most frequently used for drought assessment because of its high spatial and temporal resolution. While MODIS and Sentinel are also useful tools for tracking drought conditions, they provide free access and regular Earth observations with extensive temporal and spatial coverage (Mullapudi et al. 2023).

Satellite data are used to monitor the current situation, as well as pre- and post-disaster conditions (Belal et al. 2012). The technique of remote sensing provides quick, accurate, and continuous relevant data information about vast areas. In the present day, remote sensing-derived products, such as evapotranspiration rates, soil moisture, and vegetation indices, are available for use in agricultural drought monitoring (Rembold et al. 2015). There are a variety of archived remotely sensed data with different spatial and temporal resolution acquired by the different sensors like MODIS, Advanced Spaceborne Thermal Emission and Reflectance

Table 3 Current regional and global satellite missions relevant to drought monitoring from 1990 (prepared by author)

Satellite Mission	Launch Year	Agency/Operator	Primary Sensors	Sensor Type	Drought-Related Applications	Status
BIOMASS	2025 (Planned)	ESA	P-band SAR	Microwave	Vegetation, Biomass	Upcoming 🕒
NISAR	2025	NASA/ISRO	L- & S-band SAR	Microwave	Soil Moisture, Vegetation, Biomass	Launched
SWOT	2022	NASA/CNES	KaIn Radar	Microwave	Surface Water, Inundation	Active (✓)
Landast - 9	2021	NASA/USGS	OLI-2, TIRS-2	Multispectral + Thermal	LST, Vegetation greenness, crop health, Moisture, Water stress	Active (✓)
HySIS	2018	ISRO	Hyperspectral Imager	Hyperspectral	Hyperspectral (Crop/Soil)	Active (✓)
ICESat-2	2018	NASA	ATLAS LiDAR	Active Optical (LiDAR)	Vegetation, water storage	Active (✓)
GRACE-FO	2018	NASA/DLR	Microwave ranging	Microwave	Groundwater Storage	Active (✓)
FY-3D	2017	CMA	MERSI-2	Multispectral + Thermal	LST, Vegetation, Moisture	Active (✓)
GOES series (16, 17, 18, U)	2016-2024	NOAA	ABI	Multispectral + Thermal	Evapotranspiration, LST, rainfall	Active (✓)
SMAP	2015	NASA	L-band Radiometer/Radar	Microwave	Soil Moisture	Active (✓)
Himawari-8	2014	JAXA	AHI	Multispectral + Thermal	Vegetation, LST	
ALOS-2	2014	JAXA	PALSAR-2	Microwave	Soil Moisture, Floods	Active (✓)
Sentinel series	2014-2018	ESA	C-band SAR, MSI, OLCI, SLSTR	Microwave, Multispectral + Thermal	Soil moisture, NDVI, VCI, Crop Stress, LST, NDVI, ET	Active (✓)
GPM	2014	NASA/JAXA	DPR, GMI	Microwave	Precipitation	Active (✓)
Landsat 8	2013	NASA/USGS	OLI, TIRS	Multispectral + Thermal	NDVI, LST, NDMI	Active (✓)

Table 3 (continued)

Suomi-NPP	2011	NASA/NOAA	VIRS	Multispectral + Thermal	Vegetation health & LST	Active (✓)
Resourcesat-2 & Resourcesat-2A	2011 & 2016	ISRO	LISS-III, LISS-IV	Multispectral (Optical)	NDVI, Agriculture	Active (✓)
SMOS	2009	ESA	MIRAS	Microwave	Soil Moisture, ET	Active (✓)
RISAT-2	2009	ISRO	X-band SAR	Microwave	Soil Moisture, Inundation	Active (✓)
MetOp A/B/C (ASCAT)	2006-2018	EUMETSAT	ASCAT	Microwave	Soil Moisture	Active (✓)
CloudSat	2006	NASA	CPR Radar	Microwave	Precipitation	Active (✓)
CALIPSO	2006	NASA/CNES	CALIOP LiDAR	Active Optical (LiDAR)	Aerosols, Atmosphere	Active (✓)
Resourcesat-1	2003	ISRO	LISS-III	Multispectral (Optical)	NDVI	Decommissioned (2013)
GRACE	2002	NASA/DLR	Microwave ranging	Microwave	Groundwater Storage	Decommissioned (2017)
Aqua (MODIS/AIRS)	2002	NASA	MODIS, AIRS	Multispectral + Thermal	LST, ET, Precipitation	Active (✓)
EO-1 (Hyperion)	2000	NASA	Hyperion	Hyperspectral	Hyperspectral (NDVI, VCI)	Decommissioned (2017)
Terra (MODIS/ASTER)	1999	NASA	MODIS, ASTER	Multispectral + Thermal	NDVI, LST, ET	Active (✓)
Landsat 7 (ETM+)	1999	USGS/NASA	ETM+	Multispectral + Thermal	NDVI, LST, NDMI	Active (✓)
NOAA-15 to NOAA-20 (AVHRR)	1998 to present	NOAA	AVHRR	Multispectral + Thermal	NDVI, LST, Precipitation	Active (✓)
TRMM	1997	NASA/JAXA	PR, TMI	Microwave	Rainfall estimate	Decommissioned (2015)
ERS-1 / ERS-2	1991 & 1995	ESA	C-band SAR	Microwave	Soil moisture, hydrology	Decommissioned (2000 & 2011)

(ASTER), Advanced Microwave Scanning Radiometer – Earth Observing System (AMSR-E), Advanced Very High-Resolution Radiometer (AVHRR), Soil Moisture and Ocean Salinity (SMOS), Advanced wide Field Sensor (AWiFS), Linear Imaging Self Scanning Sensor (LISS-III), Enhanced Thematic Mapper (ETM+) etc. (Amani et al. 2021; Balsamo et al. 2018; Peng et al. 2017) are available for drought monitoring.

Remote sensing data can also be used to calculate different methods and indices to monitor agricultural drought, like NDVI, VCI, Crop Moisture Index (CMI), TCI, VHI, Soil Moisture Deficit Index (SMDI), SPI, PDSI, RAI, Surface Water Supply Index (SWSI), Drought Area Index (DAI), etc. (Balew et al. 2021; Niemeyer 2008; Zargar et al. 2011; Kaliraj et al. 2025) (refer to Fig. 5). These indices rely on distinctive spectral patterns exhibited by surface soil and canopy characteristics, specifically within the red, near-infrared, shortwave infrared, and thermal spectral ranges (Hazaymeh and Hassan 2017). For example, the Water

Deficit Index (WDI) was developed by Moran et al. (1994). Kogan (1995) proposed the VCI for drought monitoring, aiming to remove the influence of both spatial variation in the NDVI and other geographic and ecological parameters. Later, the normalized difference water index (NDWI) was proposed by Gao (1996), followed by the development of the Vegetation Temperature Condition Index (VTCI) (Wang et al. 2001). Sandholt et al. (2002) put forward the temperature vegetation drought index (TVDI) to estimate soil moisture based on the relationship between land surface temperature (LST) and vegetation index (VI); this method is significant for monitoring agricultural drought conditions through soil moisture assessment. This information was later used to propose the VHI by linearly integrating TCI and VCI (Boken et al. 2004; Kogan et al. 2013). Agricultural drought can be analyzed both spatially and temporally using remote sensing data and Geographic Information Systems (GIS) techniques. There are at least four main types of sensors that can be utilized for collecting data in remote

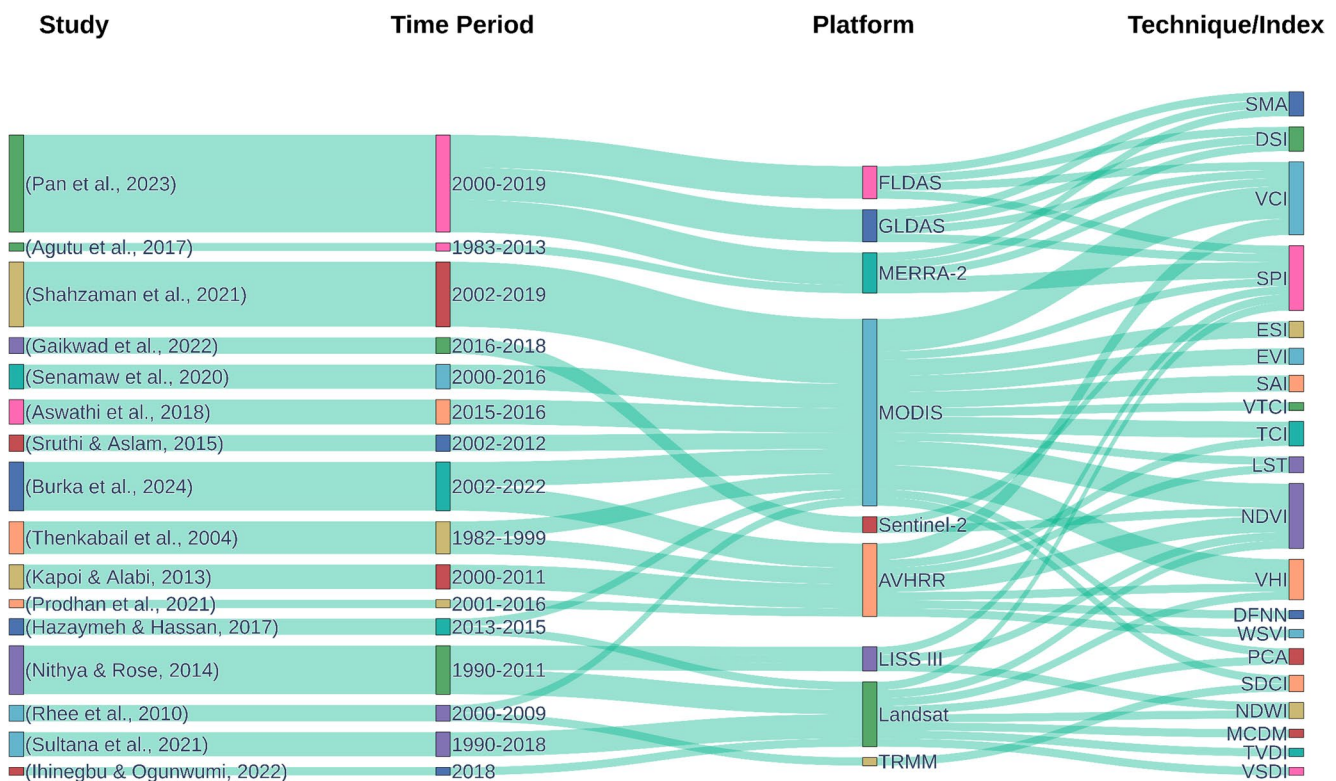


Fig. 5 Various remote sensing data sources and techniques used for agriculture drought assessment (adapted and modified from Mullapudi et al. 2023)

sensing, including optical, thermal, passive microwave, and active microwave systems. Each type has its own benefits and drawbacks (Wang and Qu 2009).

- i) For identifying vegetation conditions, soil water status, and evapotranspiration, mostly optical and thermal data were used (AghaKouchak et al. 2015; Kaliraj et al. 2022).
- ii) Microwave remote sensing has mainly been used to estimate the soil moisture content, which is the most important indicator of agricultural drought (Njoku and Entekhabi 1996).
- iii) LiDAR is the best approach to obtaining structural information of vegetation, and it can also be used to retrieve various biochemical variables, such as leaf water content (Zhu et al. 2017).

5.1 Optical Remote Sensing method for agriculture drought monitoring

Agricultural drought assessment primarily involves understanding or monitoring soil water/moisture content, as well as vegetation condition. Optical remote sensing provides a distinctive perspective by capturing electromagnetic radiation reflected from or emitted by the Earth's surface and atmosphere (Qin et al. 2021). Optical remote sensing

operates within specific spectral bands, including visible (VIS), near-infrared (NIR), and shortwave infrared (SWIR) wavelengths, for identifying the condition of vegetation greenness and wetness (Hazaymeh and Hassan 2017) (Table 4). However, index definitions and formulations were cross-referenced using the Index Database (<https://www.indexdatabase.de/>), an open-access repository of remote sensing indices that includes their equations, references, and reported applications. In addition to index descriptions, we provide a concise summary of the main strengths and limitations of drought-relevant optical indices to support the practical selection of indices. It provides valuable information about vegetation health, soil moisture, and drought stress. Healthy vegetation is characterized by its ability to absorb a substantial portion of incident visible light, particularly in the red spectrum, and reflect a significant amount of light in the NIR spectrum. Conversely, unhealthy or stressed vegetation tends to reflect more light in the visible spectrum and less in the NIR spectrum. This is because chlorophyll in plant leaves strongly absorbs visible light for photosynthesis, while leaf cell structure strongly reflects near-infrared light (Cohen-Cline et al. 2015). Surface reflectance often increases with greater water deficiency, especially in the SWIR, when studying the spectral response of plants at varying water content levels. This relationship between water content and surface reflectance can be used to assess

Table 4 Agriculture drought monitoring indices derived from optical remote sensing

Index/Method	Formula/Expression	Application	Application Area	Main strength	Key limitation	References
Soil Adjusted Vegetation Index (SAVI)	$SAVI = \frac{\rho_{NIR} - \rho_{RED}}{(1 + L) \rho_{NIR} + \rho_{RED} + L}$	Understanding drought conditions using the drought indicator	Kenya	Reduces soil background effect compared to NDVI (better in sparse vegetation)	Requires soil adjustment factor (L), not stable across all regions	(Bowell et al. 2021)
Enhanced Vegetation Index (EVI)	$EVI = G' \frac{\rho_{NIR} - \rho_{RED}}{\rho_{NIR} + C_1 \rho_{RED} - C_2 \rho_{Blue} + L}$	Drought assessment using the vegetation index	East Asia	Better than NDVI in high biomass; reduces atmospheric effects	Needs blue band; sensor dependent; more complex than NDVI	(Yoon et al. 2020)
Standardized Vegetation Index (SVI)	$z = \frac{NDVI - \overline{NDVI}}{\sigma}$ $SVI = P(Z < z)$	Agriculture drought stress assessment	East Asia	Provides a standardized anomaly-based drought signal, enabling comparison across years and regions	Requires a long-term historical reference period; sensitive to land-cover and cropping pattern changes	(Yoon et al. 2020)
Simple Ratio Water Index (SRWI)	$SRWI = \frac{\rho_{NIR}}{\rho_{SWIR1}}$	Measurement of drought events	Brazil	Simple water stress detection	Often site-specific; affected by canopy/soil mixture	(Picoli et al. 2017)
Visible and Shortwave Infrared Drought Index (VSDI)	$VSDI = \frac{1 - (\rho_{SWIR} + \rho_{RED} - 2\rho_{Blue})}{\rho_{SWIR1}}$	Application for soil and vegetation water content	Jordan	Useful to monitor both soil and vegetation drought	Sensitive to atmospheric correction and sensor differences	(Hazaymeh and Hassan 2017)
Normalized Difference Infrared Index (NDII)	$NDII = \frac{\rho_{NIR} - \rho_{SWIR2}}{\rho_{NIR} + \rho_{SWIR2}}$	Monitoring vegetation water stress	China	Strong indicator of vegetation water content, often detects stress earlier than NDVI	Sensitive to mixed pixels (soil & vegetation) and requires SWIR availability; affected by clouds/atmospheric noise	(Zhao et al. 2013)
Normalized Drought Water Index (NDWI)	$NDWI = \frac{\rho_{NIR} - \rho_{SWIR1}}{\rho_{NIR} + \rho_{SWIR1}}$	Monitoring vegetation water content for agricultural drought	India	Highly sensitive to vegetation/canopy moisture stress, useful for early drought detection	Influenced by soil background, atmosphere/cloud effects, and vegetation phenology, reducing drought-specificity	(Chakraborty and Sehgal 2010)
Modified Perpendicular Drought Index (MPDI)	$MPDI = \frac{1}{1 - FVC} (PDI - FVC * PDI_V)$	Crop water stress for drought estimation	China	Enhances drought detection by reducing the vegetation fraction effect, applicable over variable soil types and topography	Requires additional parameters/assumptions; can be unstable across crops/regions	(Ghulam et al. 2008)
Normalized Multiband Drought Index (NMDI)	$NMDI = \frac{\rho_{NIR} - (\rho_{SWIR1} - \rho_{SWIR3})}{\rho_{NIR} + (\rho_{SWIR1} + \rho_{SWIR3})}$	Application for soil and vegetation water/moisture content	China	Good at capturing both soil and vegetation moisture variations	Requires SWIR bands (limits sensors); impacted by cloud contamination	(Wang and Qu 2007)
Green Chlorophyll Index (GCI)	$GCI = \left(\frac{NIR}{GREEN} \right)^{-1}$	Study for estimating chlorophyll content		Effective proxy for chlorophyll/vigor, supports early stress detection in crops	Not drought-specific, chlorophyll decline may also be caused by nutrient stress, pests, or disease	(Gitelson et al. 2005)
Chlorophyll Red-edge Index (CI Red-edge)	$CI \text{ Red-edge} = \left(\frac{NIR}{RED_{EDGE}} \right)^{-1}$	Study for the estimation of chlorophyll content		Detects early chlorophyll stress and avoids NDVI saturation in dense vegetation	Requires sensors with red-edge bands (limited satellite availability); stress may not be exclusively drought-driven	(Gitelson et al. 2005)
Vegetation Condition Index (VCI)	$VCI = 100 * \frac{NDVI_t - NDVI_{min}}{NDVI_{max} - NDVI_{min}}$	Vegetation condition for drought monitoring	India	Good drought signal by normalizing vegetation condition relative to the historical range	Needs long-term data record; sensitive to land cover change	(Singh et al. 2003)
Normalized Ratio Vegetation Index (NRVI)	$NRVI = \frac{RVI - 1}{RVI + 1}$	For understanding the leaf area index		Related to LAI/APAR and canopy condition	Not directly a drought index; responds strongly to vegetation density, not drought only	(Baret and Guyot 1991)
Moisture Stress Index (MSI)	$MSI = \frac{\rho_{SWIR2}}{\rho_{NIR}}$	Plant water stress detection		Captures moisture stress using SWIR/NIR	May confuse drought with phenological changes; influenced by soil and canopy effects	(Hunt and Rock 1989)
Perpendicular Vegetation Index (PVI)	$PVI = \frac{1}{\sqrt{M^2 + 1}} (\rho_{NIR} - M \rho_{RED} - 1)$	Vegetation and soil condition	USA	Minimizes soil background noise	Needs soil line estimation; not robust in heterogeneous landscapes	(Richardson & Wiegand, 1977)
Normalized Difference Vegetation Index (NDVI)	$NDVI = \frac{(\rho_{NIR} - \rho_{RED})}{(\rho_{NIR} + \rho_{RED})}$	Vegetation health monitoring	US	Simple, widely used vegetation condition indicator; good long-term trend monitoring	Saturates in dense vegetation; sensitive to soil background and atmosphere	(Rouse et al. 1974)

(ρ is the surface reflectance value of blue (B), red (R), near infrared (NIR), and shortwave infrared (SWIR1, SWIR2, and SWIR3 centred at ~1.24, ~1.64, and ~2.14 μ m) bands; M is the slope of the soil line; fv is the vegetation fraction)

plant water stress and drought conditions (Holzman et al. 2021). Optical remote sensing operates within a 400–2500 nm wavelength range of the spectrum to study agricultural drought conditions. In between, water exhibits two major absorptions at 1470 nm and 1900 nm, as well as two minor peaks at 970 nm and 1200 nm (Babiet et al. 2018).

In general, the optical remote sensing-based agricultural drought indices can be further divided into three groups e.g.: (a) vegetation drought monitoring indices, such as NDVI, MSI, VCI, etc., which are more applicable for moderately to densely vegetative areas; (b) soil drought monitoring indices, like Perpendicular Drought Index (PDI), applicable for bare soil surface; and (c) vegetation and soil drought indices, such as moisture stress index (MSI), visible and shortwave drought index (VSDI), and normalized multi-band drought index (NMDI) (Hazaymeh and Hassan 2017) and again, vegetation drought monitoring indices are also further classified into (i) slope-based vegetation indices, and (ii) distance-based vegetation indices (Jackson and Huete 1991).

5.1.1 Multispectral Remote Sensing

Multispectral remote sensing is widely applied in agricultural management, including crop health assessment, irrigation and water resource monitoring, and detection of vegetation stress, by capturing surface reflectance at multiple wavelength bands (Ahmad et al. 2021). Multispectral sensors utilize a limited number of discrete spectral bands with relatively broad bandwidths across the visible, NIR, and SWIR regions, enabling operational monitoring of vegetation condition and moisture variability (Liaghat and Balasundram 2010). These bands support the derivation of widely used drought-related indices such as NDVI/EVI (vegetation greenness) and NDWI/NDMI/MSI (vegetation moisture and water stress). In recent years, multispectral drought-related indices have been extensively used to characterize and monitor agricultural drought conditions across different spatial scales (Mansour Badamassi et al. 2020; Prodhon et al. 2021). Key multispectral satellite missions commonly used in drought studies include the Landsat series, Sentinel-2, and MODIS, which provide long-term time series observations with varying spatial and temporal resolutions, making them suitable for agricultural drought monitoring (Namazi et al. 2023; Wassie et al. 2022; Widiyatmoko et al., 2021).

5.1.2 Hyperspectral Remote Sensing

Hyperspectral remote sensing is a modern and prominent technology that can be utilized for precision agriculture applications, including drought monitoring, crop

monitoring, yield estimation, water stress measurement, and crop classification (Singh and KV 2022). Hyperspectral-based methods are one way to enhance our ability to observe drought, according to recent research (Ramamoorthy et al. 2022). Hyperspectral remote sensing involves the acquisition of data across hundreds of narrow, contiguous spectral bands spanning the electromagnetic spectrum from visible light to shortwave infrared. It ranges from 400 to 2500 nm, which provides a detailed spectral signature of the surface. Unlike traditional broadband optical sensors, hyperspectral remote sensing offers the potential to more accurately identify physiological and biochemical changes in plants under water stress (Lu et al. 2020; Manjunath et al. 2011; Zou et al. 2023). Hyperspectral remote sensing data are available for agricultural studies from ground-based, airborne, and spaceborne sources (Wong et al. 2022). Many vegetation and moisture indices (e.g., NDVI, EVI, NDWI, MSI) were originally developed for multispectral sensors but can also be derived from hyperspectral reflectance. In contrast, red-edge and other narrow-band chlorophyll indices represent hyperspectral-capable indicators and are typically more sensitive to early crop stress. Therefore, a representative list of hyperspectral-derived indices is provided here for completeness (refer to Table 5).

NASA introduced the first successful Hyperion EO-1 as a hyperspectral satellite sensor into Earth's orbit in November 2000 (Lamine et al. 2019). In recent years, hyperspectral remote sensing has made significant progress in both commercial and research applications, surpassing multispectral remote sensing. This is due to the development of several hyperspectral remote sensing missions, including the Italian PRecursore IperSpettrale della Missione Applicativa (PRISMA), Germany's Environmental Mapping and Analysis Program (EnMAP), and NASA's Hyperspectral Infrared Imager (HypIRI) (Lopinto and Ananasso 2014; Singh et al. 2020).

5.2 Thermal Remote Sensing

Thermal remote sensing has several applications in agricultural drought monitoring, owing to its ability to collect temperature-related data. This approach involves assessing the temperature of both crops and soil, providing insights into water stress levels and various environmental factors that influence plant growth at different geographical scales (Hazaymeh and Hassan 2017). Thermal infrared sensors primarily detect thermal radiation emitted by the Earth's surface, and the magnitude of this radiation is associated with both LST and Land Surface Emissivity (LSE), which can serve as indicators of drought stress and plant stress (Hulley & Hook, 2011). So, temperature-based drought indices can also be calculated using thermal remote sensing data

Table 5 Agriculture drought monitoring indices derived from hyperspectral observations

Index/Method	Formula/Expression	Year	References
Normalized Difference Vegetation Index (NDVI)	$NDVI = \frac{\rho_{800} - \rho_{680}}{\rho_{800} + \rho_{680}}$	2017	(Sun et al. 2017)
Enhanced Vegetation Index (EVI)	$EVI = 2.5 \left(\frac{R_{1828} - R_{630}}{R_{1828} + 6R_{630} - 7.5R_{450} + 1} \right)$	2014	(Son et al. 2014)
Leaf Water Index (LWI)	$LWI = \frac{R_{1350}}{R_{1450}}$	2008	(Seelig et al. 2008)
Green Chlorophyll Index (Cl_{green})	$Cl_{green} = \left(\frac{\rho_{750}}{\rho_{550}} \right)^{-1}$	2005	(Gitelson et al. 2005)
Red edge Chlorophyll Index ($Cl_{red-edge}$)	$Cl_{red-edge} = \left(\frac{\rho_{750}}{\rho_{710}} \right)^{-1}$	2005	(Gitelson et al. 2005)
Simple Ratio (SR)	$SR = \frac{\rho_{800}}{\rho_{680}}$	1997	(Gitelson and Merzlyak 1997)
Normalized Difference Water Index (NDWI)	$NDWI = \frac{\rho_{857} - \rho_{1241}}{\rho_{857} + \rho_{1241}}$	1996	(Gao 1996)
Water Index (WI)	$WI = \frac{\rho_{900}}{\rho_{970}}$	1993	(Peñuelas et al. 1993)
Moisture Stress Index (MSI)	$MSR = \frac{R_{1600}}{R_{820}}$	1989	(Hunt and Rock 1989)

(ρ and R are the surface reflectance value of blue (B), red (R), near infrared (NIR), and shortwave infrared (SWIR1, SWIR2, and SWIR3 centred at ~1.24, ~1.64, and ~2.14 μ m) bands)

(Hazaymeh and Hassan 2017). AVHRR, ASTER, LANDSAT, MODIS, and ATLAS are some thermal infrared sensors deployed on satellite and airborne platforms. The most commonly used thermal remote sensing indices are given in Table 6.

5.3 Microwave Remote Sensing

Microwave remote sensing offers advantages for assessing agricultural drought, including the ability to conduct observations both during the day and at night and deeper signal penetration, which provides a more comprehensive assessment of

moisture conditions across the entire canopy (Table 7). It can be used in all weather conditions and can penetrate through clouds, which is not possible with other remote sensing techniques (Vreugdenhil et al. 2022). Microwave sensors detect signals in the frequency range of 0.3 to 300 GHz, equal to wavelengths from 1 m to 1 mm (Ulaby et al. 1986). L-, C-, X-, and Ku-bands operate using microwave sensors for earth observation, spanning frequencies between 1 and 18 GHz, and are highly sensitive to the water content in the surface soil layer (Vreugdenhil et al. 2022). Because of their sensitivity to the dielectric constant, microwaves can detect the amount of water in the top layer of soil and the biomass above ground.

Table 6 Agriculture drought monitoring indices using thermal remote sensing

Index/Method	Formula/Expression	Application	Application Area	Main strength	Key limitation	References
Temperature Ratio Index (TRI)	$TRI = \frac{\left(\frac{dT}{dt}\right)_{Max} - \left(\frac{dT}{dt}\right)_i}{\left(\frac{dT}{dt}\right)_{Max} - \left(\frac{dT}{dt}\right)_{Min}}$	Used for the analysis of agricultural drought	Australia	Simple and effective for representing relative thermal stress by comparing temperature signals	Sensitive to seasonal background temperature changes; interpretation may vary across regions and crop stages	(Hu et al. 2020)
Apparent Thermal Inertia (ATI)	$ATI = C * \frac{1-\alpha}{\Delta T_s}$	Estimate soil water content using thermal index	Italy	Strong indicator of surface moisture status, and useful for early drought stress detection	Requires day–night LST and accurate surface parameters; performance reduces under dense vegetation/clouds and with coarse temporal resolution	(Claps and Laguardia 2004)
Normalized Difference Temperature Index (NDTI)	$NDTI = \frac{T_1 - T_s}{T_1 - T_0}$	Estimation of vegetation cover and condition, as well as soil moisture, for drought circumstances.	Australia	Captures temperature anomaly/stress signal clearly; useful for identifying hot/dry conditions over croplands	Highly affected by cloud contamination, emissivity uncertainty, and surface heterogeneity (bare soil vs. vegetation)	(McVicar and Jupp 1998)
Temperature Condition Index (TCI)	$TCI = \left(\frac{T_{S_{Max}} - T_s}{T_{S_{Max}} - T_{S_{Min}}} \right) * 100$	Calculate the drought effect on vegetation	US	Useful for agricultural drought because temperature rises under water stress	Temperature can rise due to heatwaves/urbanization also and require clear sky conditions	(Kogan 1995)

(C is the solar correction factor; α is the surface albedo; ΔT_s is the difference between the afternoon and midnight land surface temperature; $T_{S_{Max}}$ and $T_{S_{Min}}$ are the maximum and minimum T_s from all images in the dataset, respectively. T_1 and T_0 are the modeled surface temperatures if there is an infinite or zero surface resistance, respectively)

Table 7 Agriculture drought monitoring indices using microwave remote sensing

Index/Method	Formula/Expression	Application	Applica- tion Area	Main strength	Key limitation	References
TRMM Precipitation Condition Index (PCI)	$PCI = \frac{TRMM - TRMM_{Min}}{TRMM_{Max} - TRMM_{Min}}$	Monitoring drought under different weather conditions	China	Provides a normalized rainfall anomaly signal, useful for the rapid detec- tion of drought onset at the regional scale	Limited by TRMM spatial resolution and retrieval uncertainty, and rainfall deficit may not directly represent agricultural drought impacts	(Zhang and Jia 2013)
Soil Moisture Condition Index (SMCI)	$SMCI = \frac{SM - SM_{Min}}{SM_{Max} - SM_{Min}}$	Monitoring short-term drought under different weather conditions	China	Directly represents soil water deficit, making it highly relevant for agricultural drought monitoring and crop stress assessment	Satellite soil moisture data are coarse and sensitive to vegeta- tion cover, surface roughness, and depth limitations, so they may not accurately reflect the moisture in the deeper root zone.	(Zhang and Jia 2013)
Microwave Polarization Difference Index (MPDI)	$MPDI = \frac{T_{BV} - T_{BH}}{T_{BV} + T_{BH}}$	Retrieving soil moisture and vegetation opti- cal depth	USA	Good soil moisture retrieval even under some vegetation	Coarse resolution in passive microwave; mixed pixel problem	(Owe et al. 2001)
Normalized Backscatter Moisture Index (NBMI)	$NBMI = \frac{B_{t1} - B_{t2}}{b_{t1} - b_{t2}}$	High spatial resolution for soil moisture conditions	Israel	Works in all weather, day/night; sensitive to soil moisture	Backscatter is affected by sur- face roughness and vegetation structure	(Shoshany et al. 2000)

$TRMM_{Min}$ and $TRMM_{Max}$; SM_{Min} and SM_{Max} are the minimum and maximum values of TRMM and SM of the pixel during the study period, respectively. T_{BV} and T_{BH} are brightness temperatures at Vertical and Horizontal polarization, respectively; b_{t1} and b_{t2} are the backscatter coefficients at different time steps

Microwaves can detect soil, vegetation, and water through their sensitivity to the dielectric constant (Wang and Qu 2009). Two types of microwave remote sensing, active and passive microwave remote sensing-based models/indices, have shown promising results for water content estimation and agricultural drought studies (Moran 2004).

Passive microwave remote sensing involves measuring the natural microwave radiation emitted by the Earth's surface and atmosphere. This remote sensing measures the soil surface emission intensity using a radiometer (Engman 1991). It can provide information about the surface temperature, vegetation cover, and soil moisture content (Njoku and Entekhabi 1996). The Scanning Multichannel Microwave Radiometer (SMMR), Special Sensor Microwave/Imager (SSM/I), and SMAP are examples of passive microwave remote sensing sensors.

Active microwave sensors are particularly useful for monitoring soil moisture and assessing the onset and severity of agricultural drought, as they can operate both day and night and have the ability to penetrate cloud cover (Walker et al. 2004). In active microwave remote sensing, sensors such as Synthetic Aperture Radar (SAR) systems transmit microwave radiation and capture backscattered signals at various frequencies, including C-, L-, and X-band. The radiation can be emitted and received by both horizontal polarization and vertical polarization, and at different angles (Vreugdenhil et al. 2022). This can be further grouped into three categories: (a) theoretical approaches (e.g., Integral equation model), (b) empirical approaches (e.g., Normalized Backscatter Moisture Index and Wetness Index), and (c) Semiempirical approaches (Hazaymeh and Hassan 2017).

5.4 Combined remote sensing-based method for agriculture drought monitoring

Monitoring agricultural drought with combined remote sensing-based approaches from multiple remote sensing sources and sensors to provide a more comprehensive and precise evaluation of drought conditions over a large area. These techniques use a variety of remote sensing technologies, including optical, thermal, and microwave sensors, to capture various environmental variables (Hazaymeh and Hassan 2017). Optical remote sensing sensors, such as multispectral and hyperspectral sensors, can be used to monitor vegetation health and land cover, while thermal remote sensing provides information on soil and crop temperatures, which indicate water stress (Gerhards et al. 2019). Combining optical and thermal data can provide more comprehensive insight into crops' or plants' responses to drought conditions (Zhang and Zhou 2015), for example, the VHI, Normalized Difference Drought Index (NDDI), VTCl, WDI, etc. (see Table 8).

On the other hand, microwave sensors, such as SAR data, can penetrate cloud cover and other weather conditions and are sensitive to soil moisture levels. Combining optical and thermal data with SAR can provide more accurate and precise information on soil moisture and vegetation health during drought. Recent advancements in microwave remote sensing enable the assessment of agricultural drought across diverse topographic and land-cover settings, using both active and passive microwave measurements. Specifically, missions such as ALOS-PALSAR, SMOS, and SMAP provide a valuable combination of passive and

Table 8 Agriculture drought monitoring indices using Optical, Thermal, and Microwave remote sensing

Type	Index/Method	Formula/Expression	Application	Application Area	Main strength	Key limitation	References
A combination of optical-based indices	Normalized Difference Drought Index (NDDI)	$NDDI = \frac{NDVI - NDWI}{NDVI + NDWI}$	Enhances the drought-capturing ability using both vegetation greenness and wetness conditions	USA	Combines vegetation & moisture sensitivity, enabling better detection of vegetation water stress than NDVI alone	Strongly affected by seasonality and land cover differences; can confuse drought with phenology/harvest changes	(Gu et al. 2007)
	Normalized Moisture Index (NMI)	$NMI = NDVI - NDWI$	Evaluate thermal and water stress of vegetation, the combination of vegetation greenness and wetness conditions	USA	Simple and effective indicator of canopy/soil moisture variation	Sensitive to soil background and atmospheric effects; performance decreases in sparse vegetation	(Jiang et al. 2006)
	Modified Vegetation Water Supply Index (MVWSI)	$MVWSI = \frac{RNDVI}{RLST^2}$	Monitoring drought conditions	China	Improved drought sensitivity by integrating vegetation response with atmospheric demand, useful for spatiotemporal drought tracking	Depends on temperature/LST quality and can be biased by cloud contamination and sensor noise	(Wu and Lu 2016)
A combination of surface temperature and vegetation Index	Crop Water Stress Index (CWSI)	$CWSI = \frac{(T_s - T_{a0}) - (T_s - T_{a0})_{Min}}{(T_s - T_{a0})_{Max} - (T_s - T_{a0})_{Min}}$	Agriculture drought monitoring	Africa	Directly represents crop water stress and irrigation requirement; highly useful at field scale	Requires local calibration and is sensitive to weather variability	(Tang and Li 2014)
	Temperature Vegetation Wetness Index (TVWI)	$TVWI = \frac{\theta_{dry} - \theta_{wet}}{\theta_{dry} + \theta_{wet}}$	Estimate land surface water content	Canada	Integrates vegetation and thermal information to reflect surface wetness, useful for early crop stress	Needs stable NDVI–LST relationship; affected by heterogeneous land cover and mixed pixels	(Hassan et al. 2007)
	Temperature Vegetation Dryness Index (TVDI)	$TVDI = \frac{T_s - T_{s_{Min}}}{T_{s_{Max}} + T_{s_{Min}}}$	Applied for the assessment of surface soil moisture status	Africa	Strong for estimating relative soil moisture/dryness using the NDVI–LST triangle method	Assumes well-defined wet/dry edges; errors in complex terrain, and irrigated areas	(Sandholt et al. 2002)
Combined optical, thermal, and microwave index	Vegetation Temperature Condition Index (VTCI)	$VTCI = \frac{T_{s_{NDVI_{Max}}} - T_{s_{NDVI_{Min}}}}{T_{s_{NDVI_{Max}}} + T_{s_{NDVI_{Min}}}}$	Drought monitoring	World	Combines greenness and temperature effects, giving a robust indicator of agricultural drought severity	Sensitive to land cover change and crop calendar; may miss short-term drought if the compositing period is long	(Kogan 2002)
	Vegetation Water Supply Index (VWSI)	$VWSI = \frac{T_s - T_{s_{Min}}}{T_s}$	Monitoring drought occurrences at the regional level	China	Effective proxy of soil moisture stress by linking NDVI with LST-based dryness conditions	Requires correct dry/wet edge definition; impacted by clouds and topographic temperature gradients	(Wang et al. 2001)
	Temperature Vegetation Index (TVX)	$TVX = \frac{T_s}{NDVI}$	Understanding drought conditions and broad scales	Africa	Captures vegetation–temperature coupling and stress response, useful for drought detection	The relationship varies by crop type and phenology stage.	(Lambin and Ehrlich 1996)
Combined optical, thermal, and microwave index	Vegetation Water Supply Index (VWSI)	$VWSI = \frac{NDVI}{T_s}$	Delineate the drought condition	USA	Links vegetation response with water supply, helpful for seasonal drought monitoring	Strongly affected by phenology and vegetation density; weaker in heterogeneous landscapes	(Carlson et al. 1994)
	Temperature Microwave Vegetation Index (TMVDI)	$TMVDI = \frac{LST_i - LST_{Min}}{a + b * MPDI - LST_{Min}}$	Monitoring a heavy drought area	China	Integrates thermal and microwave signals, improving drought detection under cloud cover and vegetation effects	Requires multiple sensors and careful preprocessing; uncertainty from microwave retrieval noise	(Liu et al. 2017)
	Microwave Integrated Drought Index (MIDI)	$MIDI = \alpha * PCI + \beta * SMCi + (1 - \alpha - \beta) TCI$	Monitoring agriculture and meteorological drought over cropland and grassland	China	Uses microwave sensitivity to soil moisture, good for drought monitoring even under clouds/haze	Microwave coarse resolution and vegetation effects limit use at the field scale and dense canopy	(Zhang and Jia 2013)
Combined optical, thermal, and microwave index	Scaled Drought Condition Index (SDCI)	$SDCI = (1/4) * \text{Scaled TS} + (2/4) * \text{Scaled TRMM} + (1/4) * \text{Scaled NDVI}$	Agricultural drought assessment over arid and humid regions	USA	Multi-indicator composite index improves reliability by combining different drought signals	Composite weights/normalization can introduce subjectivity; results depend on input datasets and scaling choices	(Rhee et al. 2010)

(VSWMin and VSWMax are the minimum and maximum values of VSWI of the pixel during the period of study; T_{Max} is the maximum surface temperature at the dry edge; T_{Min} is the minimum surface temperature at the wet edge; $T_{s_{NDVI_{Max}}}$ and $T_{s_{NDVI_{Min}}}$ are the maximum and minimum land surface temperatures of pixels that have the same NDVI value in a study area, respectively. $T_{s_{NDVI}}$ is the land surface temperature of one pixel whose NDVI value is NDVI; T_a is the air temperature; ET is the actual evaporation, θ_{dry} is the dry edge; θ_{wet} is the wet edge; θ_s is the surface potential temperature, LST_i means surface temperature, LST_{Min} lower surface temperature limit, a and b represent undetermined coefficients, MPDI means Microwave Polarization Difference Index.)

active microwave data, and they can provide highly accurate drought monitoring solutions by improving the accuracy of soil moisture and vegetation water content retrievals (Hazaymeh and Hassan 2017).

6 Big data and machine learning (ML) methods for agricultural drought study

Big data and ML have become increasingly important for agricultural drought monitoring, as they can handle large, heterogeneous datasets and extract insights that enable real-time analysis, predictive modeling, and improved decision-making. As droughts grow more frequent and severe worldwide, interest in integrating big data and ML techniques has intensified, engaging engineers, environmental scientists, and researchers from diverse disciplines (Prodhan et al. 2022). Because manual processing and interpretation of remote sensing imagery and meteorological data are labor-intensive and time-consuming, big data and machine learning offer effective, scalable alternatives for feature extraction and information retrieval from large spatiotemporal datasets. Within computational intelligence, these approaches have become essential for drought monitoring and prediction, improving the accuracy of both situational assessment and forecasting. In particular, the remote sensing data cubes, which organize Earth observation and ancillary data into a consistent spatiotemporal-spectral array, enable efficient, reproducible queries and large-scale analytics, further streamlining drought assessment and forecasting workflows (Xu et al. 2020).

Big data analytics allow for processing large datasets from multiple sources, like remote sensing, meteorological stations, and agriculture surveys, to track drought conditions adequately (Balti et al. 2020). For example, Zhang et al. (2017) analysed and predicted a big data-based approach for the California drought. The authors used diverse datasets, including climate sensor and satellite data, weather data, drought conditions, and water-use reports. Similarly, Shah et al. (2017) proposed a Big Data analytics approach to predict drought occurrence using a Random Forest (RF) model and manage droughts based on historical rainfall, temperature, and evapotranspiration data.

Similarly, Machine Learning and Deep Learning models have been found helpful in evaluating complex datasets and detecting patterns associated with drought by using past data (Agana and Homaifar 2017; Prodhan et al. 2021, 2022; Rahmati et al. 2020). Popular ML models in the literature include decision tree models, such as Classification and Regression Trees (CART) (Veetil and Mishra 2023), Conditional Inference Trees and multivariate regression trees (Paez-Trujillo et al. 2023), Support Vector Machine (SVM) models (Linear

SVM (Pande et al. 2023), Kernel SVMs (Pande et al. 2023), Support Vector Regression (Tian et al., 2018) and One-Class SVM (Roodposhti et al. 2017), Boosting models (Adaptive Boosting (AdaBoost) (Koteswararao and Rajendran 2025), Gradient Boosting Machines (GBM), Xtreme Gradient Boosting (XGBoost) (Ekmekcioğlu 2023), Categorical Boosting (CatBoost), Ensemble models (RF) (Zarei et al. 2023), Bagging models, Extremely Random Trees (XTR) (Elbeltagi et al. 2023), Stacking models) and Neural network-based models (Multilayer Perceptron (MLP) NNs (Kan et al. 2023) and deep learning models such as Convolutional Neural Networks (CNN) (Kan et al. 2023), Autoencoders (Xing et al. 2025), Long Short-Term Memory (LSTM) (Kheyruri et al. 2023), and Graph Neural Networks (GNN) (Chandra et al. 2025) have been widely used to predict agricultural droughts with high accuracy. Although picking a single most suitable model still remains a topic of debate.

Machine learning techniques are coupled with remote sensing data, such as satellite imagery and climate data, to improve the temporal and spatial accuracy of drought monitoring. Rahmati et al. (2020) evaluate the Relative Departure Soil Moisture (RDSM) method for mapping agricultural drought hazards using CART, boosted regression trees (BRT), RF, multivariate adaptive regression splines (MARS), flexible discriminant analysis (FDA), and SVM in Australia. The study found that approximately 26% of the area is at high or very high risk of drought. Similar work was conducted by Kafy et al. (2023) using Cellular Automata (CA) and Artificial Neural Network (ANN) algorithms to assess and predict agricultural drought vulnerability in Bangladesh, based on remote sensing-based indices. Similarly, Nie et al. (2018) employed SVM, RF, ANN, and BPNN (Back-Propagation Neural Network) models to predict soil moisture. The findings indicated that SVM outperformed RF and BPNN in terms of prediction accuracy, with terrain, precipitation, and relative humidity being the primary factors influencing soil moisture. Therefore, big data integration approaches would benefit real-time drought disaster forecasting and mapping. The importance of environmental decision-making in drought disaster analysis will be further enhanced by a knowledge-based system with a big data platform, incorporating ML and data mining (Prodhan et al. 2022). However, there is still a need for different hybrid and physics based ML techniques such as Autoencoder coupled with RF/XGBoost, Physics-Informed Neural Network (PINN), Process-Guided Neural Networks (PGNNs) with high computational efficiency that will likely focus on leveraging large amounts of data sets (big data), such as combining remote sensing data, weather data, and other relevant data to improve the accuracy and speed of predictions. Table 9 lists contemporary models and methods for handling various features of big data in drought monitoring.

Table 9 Big data drought monitoring tools. (source: Balti et al. 2020)

Characteristics	Tool used	Remarks	References
Volume	Hadoop	Distributed processing	(Rathore et al. 2015; Zou et al. 2018)
Variety	Hadoop	Schema-less data store	(Rathore et al. 2015; Zou et al. 2018)
Velocity	Weka, H2O	Query data inside RAM instead of disk	(Yu et al. 2019; Zhang et al. 2017)
Veracity	Heroku	Stream processing	(Zhang et al. 2017)
Visualization	ArcGIS, Weka	Friendly user interface	(Xu et al. 2016)

7 Agriculture drought modeling and prediction using remote sensing data

Agricultural drought can affect multiple dimensions of society, including a decline in crop and livestock productivity. Although drought cannot be fully prevented, its impacts can be reduced through effective monitoring and early warning systems based on historical trends and predictive modelling (Ortega-Gaucin et al. 2016; Hao et al. 2014). Forecasting of drought is a challenging task due to the highly complex interactions of meteorological and hydrological processes (Wu 2014). Even agricultural drought models that rely on meteorological factors often struggle to provide precise predictions due to the alterations in established patterns caused by climate change (Park et al. 2019). Therefore, accurate agricultural drought prediction can help farmers adjust planting schedules, select suitable crops, and adapt irrigation methods, particularly in areas vulnerable to drought. In this case, remote sensing (RS) data are reliable and efficient for agricultural drought studies, as they provide spatially continuous and temporally consistent observations over large areas (Zhao et al. 2019). Satellite products capture various hydroclimatic factors relevant to drought evolution, including precipitation, surface temperature, soil moisture, snow cover, and vegetation health, on both regional and global scales (AghaKouchak et al. 2015; Wardlow et al. 2012). Traditional drought assessment approaches that depend mainly on sparse station observations, while RS data can support near-real-time monitoring with greater spatiotemporal detail (Sardar et al. 2021). This helps in understanding the onset, duration, severity, and recovery of drought.

Agricultural drought monitoring, evaluation, and forecasting are commonly based on meteorological indices (e.g., SPI, SPEI, PDSI) and vegetation/temperature-based indices (e.g., NDVI, VCI, VHI, CMI, crop-specific indices, etc.), and among other composite indices (Carlson et al. 1994; Chere and Debalke 2024; Docheshmeh Gorgij et al. 2022; Kan et al. 2023). These indices serve as valuable tools for quantifying the impacts of drought on agriculture, particularly through changes in vegetation condition

and crop stress (Park et al. 2019). Vegetation indices (VIs) are especially useful because they capture spectral changes associated with reduced greenness and moisture stress during drought periods (Lekakis et al. 2022). While meteorological indices remain widely used, satellite time-series indicators are increasingly supporting improved prediction performance by better representing spatial variability and showing strong potential for enhancing prediction accuracy (Sardar et al. 2021).

In general, drought prediction approaches can be categorized into Stochastic time-series models (AutoRegressive Integrated Moving Average (ARIMA), Seasonal Autoregressive Integrated Moving Average (SARIMA) that capture temporal persistence and seasonality), dynamic models (Global Circulation Model), index-based approaches (Index-based class), and data-driven ML/DL models that learn nonlinear relationships from multi-source predictors (Box 2013; Breiman 2001; Park et al. 2019; Sardar et al. 2021; Tian et al. 2013, 2016) (Table 10). When integrated with GIS-based spatial analysis, these approaches enhance agricultural drought early warning systems by improving predictive skills and supporting informed decision-making across various spatial scales (Ahmad et al. 2010; Chen et al. 2012; Moody and Darken 1989; Tian et al., 2018).

8 Discussions

Numerous recent studies have explored historical drought events worldwide. Mitigating the effects of drought and other extreme weather events requires timely and reliable information, which is crucial for decision-making (Mullapudi et al. 2023). Recent developments in geospatial technology and data computing have led to a notable increase in research focusing on the multivariate monitoring and modeling of agricultural drought. This comprehensive study examines the recent progress in this field, encompassing the application of multisource remote sensing data, various machine learning models, and big data approaches. Agricultural drought modelling essentially encompasses three significant components: descriptive modelling of drought characteristics (onset, intensity, and termination), multivariate predictive modelling of drought intensity, and multivariate risk mapping (Houmma et al. 2022).

A diverse range of multi-sensor approaches to agricultural drought modeling have been developed and tested around the world based on the integration of earth observation data with other data sources using various machine learning applications (Kan et al. 2023; Qin et al. 2021; Rahmati et al. 2020; Sandeep et al. 2021). Different researchers have employed various remote sensing-based indicators to monitor and analyze agricultural drought in their respective

Table 10 Shows selected studies that utilised various drought prediction models

Algorithms	Model/Technique	Application	Drought Index used	Application Area	Accuracy of the Models	Main strength	Key limitation	Reference
Hybrid Model	Empirical Mode Decomposition-Adaptive Neuro-Fuzzy Inference System (EMD-ANFIS)	Prediction of meteorological drought indices	self-calibrated Palmer Drought Severity Index (sc-PDSI)	Turkey	NSE = 0.96, MSE = 0.18	Decomposes non-stationary drought time series into simpler components, thereby enhancing the stability of ANFIS predictions	High computational cost and results depend strongly on decomposition settings (model selection)	(Basakın et al. 2021)
	Support Vector Machine-Dragonfly Algorithm (SVM-DA)	Agriculture drought prediction	PDSI	Iran	RMSE = 0.817, NRMSE = 0.097, WI = 0.940, R = 0.889	Metaheuristic tuning improves SVM parameter selection, often achieving higher accuracy than the default SVM.	Optimization can be slow/unstable and may overfit if the validation strategy is weak	(Aghelpour et al. 2021)
	Hybrid SVR with Particle Swarm Optimization (PSO) and Harris Hawks Optimization (HHO) algorithm	Predict meteorological drought	EDI	India	RMSE = 0.535–0.965, MAE = 0.363–0.622, NSE = 0.558–0.860, COC = 0.760–0.930, WI = 0.862–0.959	Strong for nonlinear drought forecasting; hybrid optimizers improve SVR generalization and reduce manual tuning	Computationally heavy; performance sensitive to optimizer hyperparameters and search space	(Malik et al. 2021)
	ANN-Pelican optimization algorithm (POA), ANFIS-POA, and SVM-POA	Agriculture drought prediction	ASPI	Iran	RMSE = 0.012, MAE = 0.178, BIAS = -0.069, NSE = 0.804, WI = 0.780, CI 0.627	Automatic hyperparameter optimization improves model performance and reduces subjective tuning	Metaheuristic optimization may converge inconsistently and increase model complexity/replicability issues	(Rezaei et al. 2025)
Machine Learning (ML)	Ensemble ML	19 individual ML models with a stacking ensemble approach to predict	SPEI	China	The CBR Model has higher accuracy with R ² = 0.9065	Improves robustness and accuracy by combining multiple learners and reduces single-model bias	Increased complexity makes interpretation difficult and may lead to over-optimization without real physical meaning	(Xu et al. 2024)
	Decision Tree (DT)	Combines 6 ML models with ensemble methods (Bagging and Voting)	Drought detection	Pakistan	The voting ensemble outperformed with the highest F1 score of 84.80%, MAE = 11.95, R ² = 0.59			(Raza et al. 2025)
		Prediction of agricultural drought indicators	PDSI	Sweden		Highly interpretable, useful for identifying key drought drivers and thresholds	Single trees easily overfit, leading to unstable drought predictions across regions	(Kan et al. 2023)
		Hydrological drought prediction	Standardized rainfall/runoff Index (SRI)	Algeria	RMSE = 0.28, MAE = 0.19, NSE = 0.86, R ² = 0.90			(Achite et al. 2022)
Random Forest (RF)		Drought Prediction	SPI	India	R = 0.961, MAE = 0.361, RMSE = 0.538	Strong performance with nonlinear relationships, handles mixed predictors, and provides feature importance	Less effective for long lead-time forecasting, and interpretability is still limited compared to simple models	(Elbeltagi et al. 2023)
		Agriculture and Hydrological Drought Prediction	SPI	Syria	R = 0.91, MAE = 0.30, RMSE = 0.42, RAE = 36.45, RRSE = 40.29			(Mohammed et al. 2022)
		Predicting drought severity sites	SMI	Kenya	RMSE = 0.382, MAE = 0.375, R ² = 0.58			(Park et al. 2019)
		short-period drought prediction	VSDI with MJO and VSDI without MJO	East Asia	R = 0.70 RMSE = 0.133			(Park et al. 2018)
Support Vector Machine (SVM)		Drought forecasting	SPI	Algeria	CC = 0.908, MAE = 0.305, RMSE = 0.431, RAE = 37.412, RRSE = 41.963	Strong performance with limited samples and high-dimensional drought predictors	Sensitive to kernel/parameter choice; weak scalability for very large spatiotemporal datasets	(Achite et al. 2023)
Artificial Neural Network (ANN)		Agriculture drought prediction	SPEI	China	MAPE = 0.218, RMSE = 0.275, NSE = 0.886			(Tian et al., 2018)
		Drought forecasting	SPI	Iran	RMSE = 0.08125, R = 0.9237	Learns nonlinear relationships between drought indices and multi-source predictors	Needs large training samples and careful regularization; prone to overfitting	(Mokhtarzad et al. 2017)
		Drought Prediction	EVI	Australia	MAE = 0.602, RMSE = 0.172, CD = 0.578, WI = 0.92			(Deo and Sahin 2015)

Table 10 (continued)

Algorithms	Model/Technique	Application	Drought Index used	Application Area	Accuracy of the Models	Main strength	Key limitation	Reference
Deep Learning	Long Short-Term Memory (LSTM)	Agriculture drought forecasting	SPI, SPEI, PDSI, NDVI/VCI	India	$R=0.82$, $MAE=0.19$, $RMSE=0.33$	Best suited for drought time-series forecasting because it learns long-term dependencies and lag effects	Requires substantial data and computing resources; difficult to interpret and sensitive to sequence design	(Chandra et al. 2025)
		Drought forecasting	SPEI, NDVI, and meteorological data	Iran	$R=0.92$, $MAPE=9.246$, $RMSE=0.03$, Bias = 0.00512			(Abbes et al. 2023)
	Convolution Neural Network (CNN)	Agriculture drought prediction Drought prediction	NDVI NDVI, SAVI, EVI, and Atmospherically Resistant Vegetation Index (ARVI)	India Ethiopia	improved accuracy up to 96% Overall accuracy of all indices is 0.89.	Excellent for extracting spatial drought patterns directly from RS imagery	Needs large labelled datasets; weak for pure temporal forecasting unless coupled with sequence models	(Sardar et al. 2021) (Chaudhari et al. 2021)
Fuzzy/Neuro fuzzy	Deep Multilayer Perceptron (MLP)	Predict long-term drought for drought monitoring	Standardized Streamflow Index (SSI)	USA	$R^2=0.91302$, $MSE=0.00303$, $RMSE=0.05507$, $MAE=0.03969$	Strong baseline deep model for multivariate drought prediction with RS and climate predictors	Easily becomes a black-box, making scientific justification difficult for drought decision-making	(Agana and Homaifar 2017)
		Drought forecasting	SPI	Malaysia	$R=0.94$, $MAE=0.28$, $RMSE=0.34$			(Hong and Hong 2015)
	Fuzzy Logic	Drought index prediction	SPI	India	Prediction accuracy is higher for all time scales	Handles uncertainty well and produces human-readable drought rules (useful for early warning)	Membership rule design is subjective, with limited scalability for large-scale, real-time drought analytics	(Malik et al. 2020)
Statistical/Time-series		Long-term drought forecasting	PDSI, Palmer Modified Drought Index (PMDI)	USA	$NSSS=0.903$, $CC=0.951$			(Özger et al. 2012)
	Seasonal Autoregressive Integrated Moving Average (SARIMA)	Agriculture drought forecasting	VTCT	China	$RMSE=0.0818$	Strong for seasonal drought indices (SPI/SPEI) time series with clear periodicity	Performance drops when drought patterns become non-stationary due to climate variability/change	(Tian et al. 2016)
	Autoregressive Integrated Moving Average (ARIMA)	Drought forecasting in a hyper-arid climate Drought forecasting	SPEI SPI	Saudi Arabia China	$R^2=0.974$, $RMSE=0.180$, $MAE=0.121$ $R=0.985$	Strong baseline for linear drought index forecasting (SPI/SPEI) using past values	Weak under nonlinear drought behavior and cannot integrate multi-source RS/climate drivers well	(Mossad and Alazba 2015) (Han et al. 2013)

study areas (Aswathi et al. 2018; Hazaymeh and Hassan 2017; Naskar 2022; Tang and Li 2014). The integration of multiple drought indices or CDI enhances the ability to accurately detect both global and local drought events. CDI is a technique that integrates multiple variables and aims to reconstruct quantitative and qualitative information for assessing and monitoring drought. Bayissa et al. (2022) developed an agricultural combined drought index (agCDI) using a principal component analysis (PCA) approach to monitor and characterize the spatial and temporal patterns of drought in Sri Lanka, employing satellite-based indices to mitigate the adverse impacts in the area.

Similarly, Chattopadhyay et al. (2020) utilized meteorological, land-based, and remote sensing observation data to develop CDI using a weightage overlay analysis in India. Other CDIs, such as the Vegetation Drought Response Index (VegDRI), were developed by Brown et al. (2008). Rojas (2021) utilized the Agricultural Stress Index System (ASIS), while Liu et al. (2020) developed an ANN-based Integrated Agricultural Drought Index (IADI), among other methods, to identify areas vulnerable to agricultural drought. Even though many researchers are now using multiple ML techniques to predict drought using various hydroclimatic and RS data, this suggests that multivariate composite models can provide valuable insights into different forms of drought simultaneously. The potential disadvantage of these models is that they require multiple datasets, which often have quite different characteristics, necessitating normalization and downscaling treatments (Houmma et al. 2022). On the other hand, multiple dimensions of agricultural drought have been described and/or simulated using both machine learning and deep learning models.

The application of ML, hybrid models, and big data is an increasingly popular trend in drought modelling. Every study employs a distinct technique in terms of the choice of response variable and the algorithm used to assess the relative importance of variables or indices in the multivariate models developed by Houmma et al. (2022). Multivariate predictive analyses offer a practical approach to mapping expected risks, including their intensity and spatial distribution (Rahmati et al. 2020). A significant number of ML/DL algorithm-based models have been used (Aghelpour et al. 2021; Fung et al. 2020; Tian et al., 2018). Bali and Singla (2021) recently demonstrated considerable support for the movement toward increased use of hybrid models and deep learning for prediction, particularly of crop yields. To address this, Tian et al. (2018) analyzed the relation between soil moisture and drought, and predicted agricultural drought in the Xiangjiang River basin, China. They developed an SVR model to predict agricultural droughts using climate indices. Simultaneously, LSTM has been utilized by Kheyruri et al. (2023) based on large-scale indicators (big data), such as

El Niño-Southern Oscillation (ENSO), NDVI, and meteorological parameters, at time lags of 1–4 months to understand the predictors of agricultural drought. They revealed NDVI values predicting better performance in semi-arid and arid areas of Iran. Similar work has been conducted (Sardar et al. 2021) to predict drought or no-drought conditions using a CNN model based on NDVI from satellite images in Karnataka, India, achieving 96% accuracy. Additionally, numerous scientific studies have been conducted to compare various types of ML and DL models for analyzing agricultural drought. For example, Kan et al. (2023) compared seven ML/DL models, namely RF, decision tree (DT), multivariate linear regression (MLR), SVR, ARIMA, ANN, and CNN, in combination with the indicators of soil moisture and PDSI. The best-performing models for predicting soil moisture were the multi-feature RF model (better suited for national-scale prediction) and the temporal ARIMA model (best suited for local-scale prediction), and all these models performed better than PDSI. In another work, the applicability of four ML algorithms, such as bagging, random subspace (RSS), random tree, and RF, has been used by Mohammed et al. (2022) in predicting drought events in the eastern Mediterranean based on SPI-3 and SPI-12. They obtained results from the bagging algorithm, which outperformed the other algorithms and is more dynamic in capturing drought. Roushangar et al. (2021) suggested that integrated models might perform noticeably better than single models, improving predictive modeling accuracy by up to 40% and 50%. However, deep learning models for drought prediction still face challenges, particularly in delivering accurate long-range forecasts and high-resolution spatial outputs, and in their reliance on large amounts of data, which are often insufficient in many regions of the world (Gyaneshwar et al. 2023).

To overcome the problem, large, long-term datasets (big data) can be used for agricultural drought analysis, especially in combination with hybrid and physics-based machine learning and deep learning approaches, leveraging higher-temporal- and spatial-resolution satellite data. Further, the emerging technologies such as UAV-based high-resolution data, high-precision field-based sensors, affordable cloud computing platforms, and integration through Internet of Things (IoT) have enabled researchers to monitor agricultural drought spatiotemporally at finer resolution and to develop an early warning system (Anandan et al. 2024; da Silva et al. 2025; Dahir et al. 2023; Fei et al. 2023; Masupha et al. 2025; Yang et al. 2024). Further, Artificial Intelligence (AI) could be explored to enhance the accuracy of agricultural predictive modelling, as it resembles biological processes and operates on logical systems such as sentential and predicate logic (Kikon and Deka 2022; Oyarzabal et al. 2025). Moreover, these

advancements are directly linked to the achievement of the United Nations SDGs, particularly SDG 2 (Zero Hunger) through safeguarding food security, SDG 6 (Clean Water and Sanitation) by ensuring sustainable water management, SDG 13 (Climate Action) by enhancing resilience against climate-related hazards, and SDG 15 (Life on Land) through the promotion of sustainable agriculture and the prevention of land degradation (Efremova et al. 2023; Jain and Mitra 2025; Mandal et al. 2025; Nan et al. 2025; Zaigham Abbas Naqvi et al., 2025; Ziesche et al. 2023).

9 Conclusion

Agriculture is the most sensitive and vulnerable sector, and it is strongly affected by rainfall variability and drought. Agricultural drought monitoring and modelling remain challenging because drought is driven by complex interactions among hydroclimatic, environmental, and human factors, which ultimately influence ecosystems, hydrology, socioeconomics, and food security. Although drought impacts cannot be fully avoided, improved monitoring and early warning can substantially reduce losses. But the inconsistent high spatial resolution and long-term data are the major obstacles. This paper provides a critical review of the details of various agricultural drought indices and techniques for monitoring and modeling using multi-source remote-sensing satellite data and ML- and DL-based geospatial approaches. This paper outlines the fundamental concepts of drought, its various types, and their impact. The review identifies that among various remote sensing spectral indices, NDVI, SMA, VCI, and VHI are widely used to characterize agricultural drought. Similarly, precipitation-based indices, such as the SPI and the SPEI, are among the most widely used indicators for drought assessment and prediction. We further found that the MODIS and Landsat datasets are the most widely used remote sensing datasets for drought assessment due to their long-term continuity and accessibility.

Beyond summarizing the existing literature, this review highlights that the choice of drought indices and satellite data is highly context-dependent, and that multi-index integration is consistently more robust than single-index drought mapping. In drought modeling and forecasting, methods such as RF, ANN, SVM, LSTM, CNN, ANFIS, and MLP are widely adopted, but their performance often depends more on input feature selection, lag design, validation strategy, and spatial transferability than on the algorithm itself. The literature also indicates that drought research increasingly covers both static drought classification and advanced probabilistic, predictive, and near-real-time early warning systems supported by ML and DL approaches. Future

research should prioritize high-resolution soil moisture estimation, especially for root-zone conditions, as it plays a crucial role in agricultural drought monitoring and crop yield assessment. The development of advanced and innovative Physics-guided and hybrid ML approaches is also necessary to improve generalization and reliability under climate non-stationarity. Additionally, it should focus on integrating drought monitoring with impact assessment, including crop yield-loss modeling and vulnerability/risk mapping.

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